



OPERATING SYSTEM DESIGN

S Thenmozhi

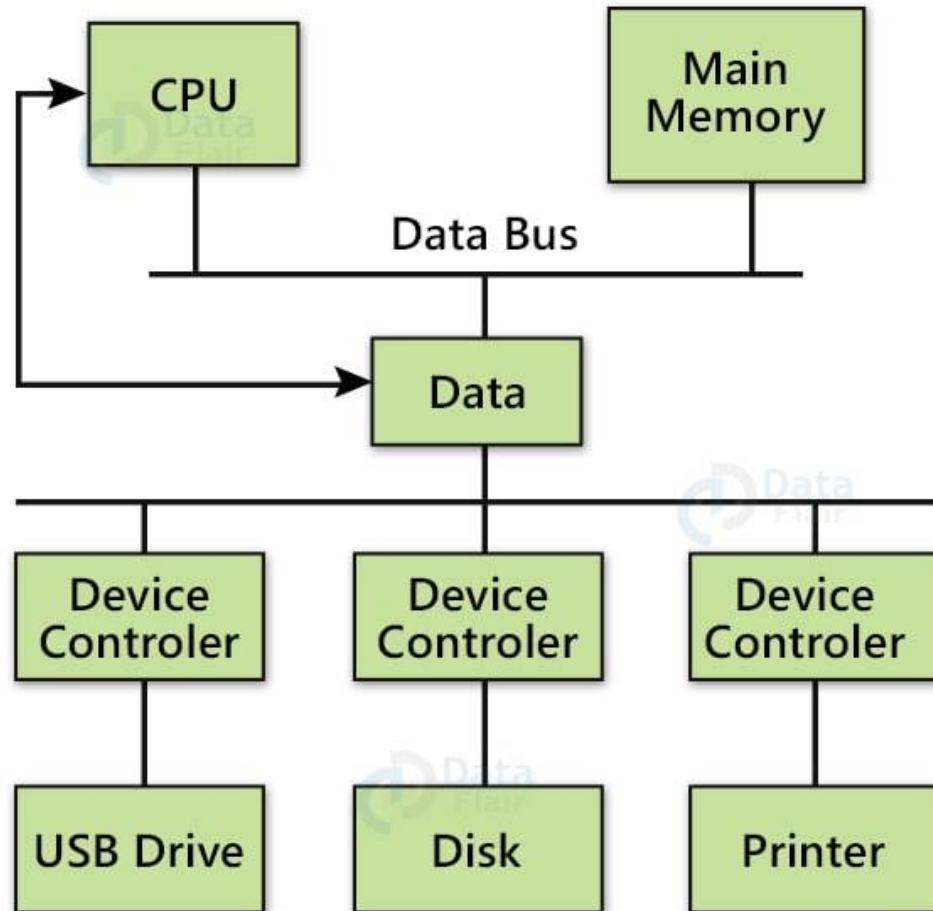
Department of Computer Applications

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Storage Management & System Security

S Thenmozhi

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Device driver or **Software Driver** is a computer program allowing high level computer programs to interact with the specific hardware device

A driver typically communicates to the **hardware device through the computer bus and communicates to other subsystems** to which the hardware connects

Drivers are **hardware dependent and OS- specific**

Device Driver is part of OS kernel. It is compiled by the OS and dynamically loaded into the OS while execution.

Device Drivers are translators between the hardware device and the applications which uses it

They help your system understand and communicate with different hardware like printers, keyboards, and monitors

They comprise the **kernel routines which is responsible for doing a specific activity through the device**

Device Driver Programs need to be allocated with memory when the device gets attached to the system. This is done with **kmalloc()** function. It inturn invokes the **get_free_page()**

Memory is freed using **kfree()** when the device is detached.

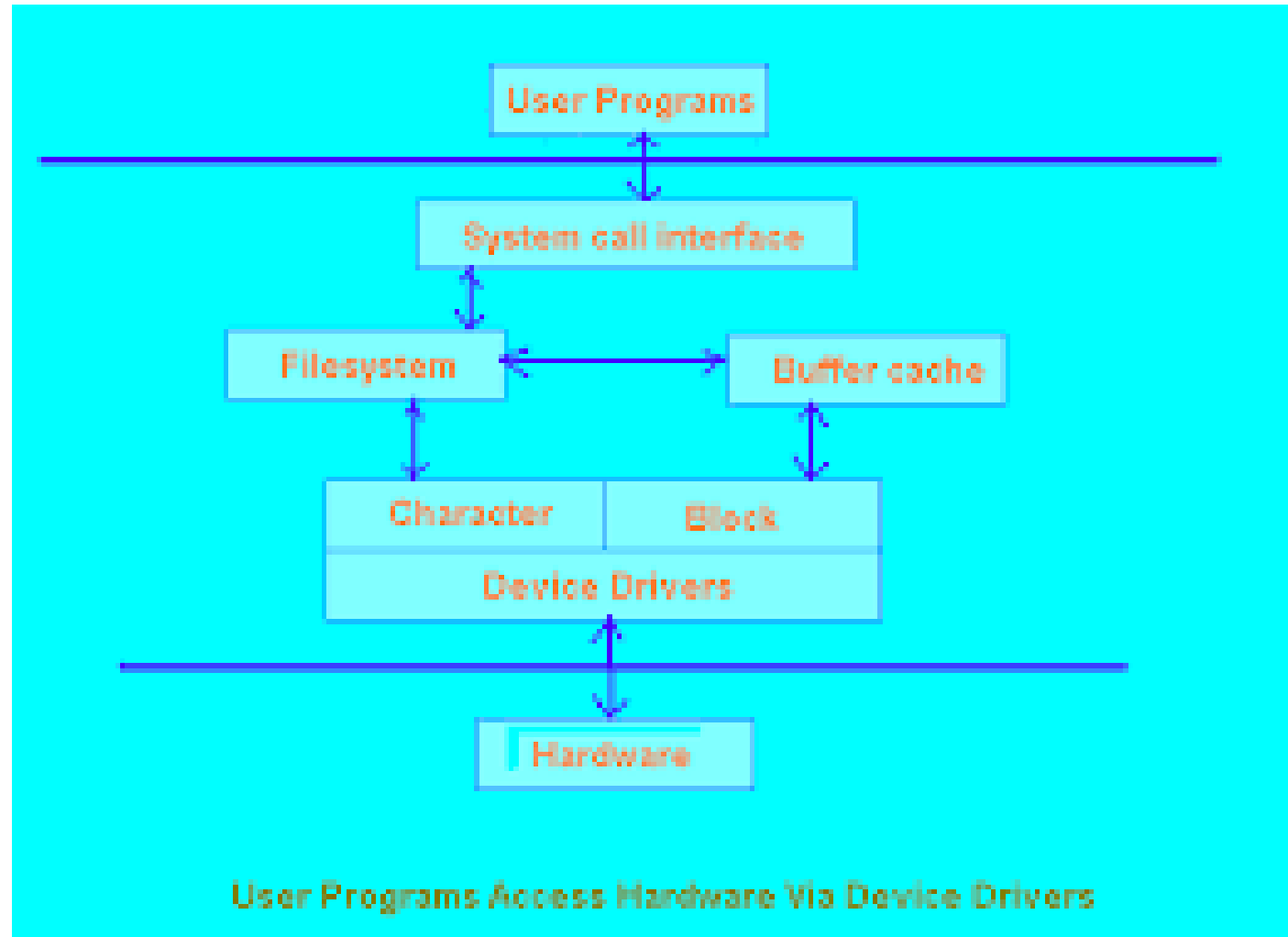
Once the device driver gets loaded, the system calls can be initiated to operate on the device.

Now, Device drivers are interfaces to the hardware

These interfaces can be **Block Device Interface (BDI)** or **Character Device Interface (CDI)**

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Device Drivers



Functions of Device Drivers

Accepts read or write requests from the application layer

Initialize the device

Manage power requirements and log events

Check input parameters are valid

Translate valid input into concrete terms

Check the device is in use?

Control the device by issuing the sequence of commands. The driver determines what commands to be issued.

Device Driver Protocol

After driver knows which commands to issue, it starts to write them into controller's device registers

After writing each commands, it check to see if the controller accepted the commands and is ready to receive the next.

After commands is issued, either the device executes the command with no delay or the controller block the commands if the device is busy and unblocks when it has become free by raising an interrupt.

Types of Devices Drivers: Character Device drivers, Block Device Drivers.

Character Devices

Character devices are like instant messengers for your computer.

They send and receive data one character at a time eg: Keyboard, mouse

Also called as Raw IO or direct IO.

The kernel does not buffer raw IO operations. It directly transfers the data between the physical memory and its buffers

These are very fast transfers

They can be accessed only sequentially

These devices use the DMA controllers (which can steal a CPU cycle and do its operation)

Two major operations: read(), write()

Devices: serial ports, parallel ports, sound cards

Block Devices

Block devices are more like delivery trucks for your data.

They handle larger chunks of information in blocks, making them efficient for tasks like reading and writing files on disks or flash drives.

Block devices support byte streams

Data sent from one place in memory to another must be broken into bytes before being transferred across an operating system's memory space

The block devices can host a file system

The read and write operations also allowed here and holds every property as the character devices

The difference is only how internally they operate. For a user, it makes no difference.



THANK YOU

S Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393